Software Requirements Specification (SRS)
Student Login Management System

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1. Introduction

The **Software Requirements Specification (SRS)** document provides a comprehensive and detailed description of the requirements for the **Student Login Management System (SLMS)**. It serves as a foundational reference for the design, development, testing, deployment, and maintenance of the system. This document is intended to ensure a shared understanding of system functionalities and constraints among all stakeholders involved in the project.

The SLMS is designed to address the need for a secure and efficient mechanism for managing student authentication and access control within an academic environment. With the increasing reliance on digital platforms in education, there is a critical need for robust systems that safeguard user data while providing seamless access to academic resources. This document outlines both the functional and non-functional aspects of the system, ensuring that all requirements are clearly defined and verifiable.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed and structured description of the requirements for the Student Login Management System (SLMS). The system is intended to manage student authentication, authorization, and secure access to an academic platform.

This document serves multiple purposes:

- It acts as a **communication tool** between stakeholders, including developers, system analysts, project managers, and clients
- It provides a **clear and unambiguous definition** of system requirements to guide development
- It establishes a **baseline for system validation and verification**, ensuring that the final product meets user expectations
- It supports **project planning, cost estimation, and risk management**

The SRS defines:

- Functional requirements that describe what the system should do
- Non-functional requirements that define system performance, usability, reliability, and security
- System features and operational behaviour
- Design constraints and assumptions
- User interactions and workflows
- Security requirements and data protection measures

By documenting these elements, the SRS ensures consistency throughout the software development lifecycle and reduces the risk of misinterpretation or incomplete implementation.

1.2 Scope

The Student Login Management System (SLMS) is a web-based application developed to provide secure and efficient authentication services for students accessing an academic platform. The system focuses on managing user identity, controlling access to resources, and ensuring data security.

The SLMS acts as a gateway through which students must pass before accessing protected academic services such as course materials, results, and institutional resources. It ensures that only authorized users can access the system by implementing strict authentication and authorization mechanisms.

The system provides the following core functionalities:

- **Student Account Registration**
  Allows new users to create accounts by providing personal and academic details
- **Secure Login Authentication**
  Verifies user identity through credentials such as email and password
- **Password Recovery and Reset**
  Enables users to recover or reset forgotten passwords securely
- **Email Verification**
  Confirms user identity during registration using verification links or codes
- **User Session Management**
  Controls login sessions, including session timeouts and logout functionality
- **Student Profile Management**
  Allows users to view and update their personal information
- **Login Activity Monitoring**
  Tracks login attempts and system access for auditing and security purposes
- **Administrative Control**
  Provides administrators with tools to manage user accounts, monitor activity, and enforce policies

The system is designed to operate within a secure environment using modern web technologies and security protocols. It ensures confidentiality, integrity, and availability of user data while maintaining ease of use.

1.3 Definitions, Acronyms, and Abbreviations

This section provides definitions and explanations of key terms used throughout the document to ensure clarity and consistency.

- **SLMS (Student Login Management System):**
  The system designed to manage student authentication and access control
- **SRS (Software Requirements Specification):**
  A document that describes the functionality and constraints of a software system
- **UI (User Interface):**
  The visual components through which users interact with the system
- **DB (Database):**
  A structured collection of data used to store system information

- **OTP (One-Time Password):**
  A temporary code used for secure authentication
- **HTTPS (Hypertext Transfer Protocol Secure):**
  A secure communication protocol used for transmitting data over the internet
- **Admin (Administrator):**
  A user with elevated privileges responsible for managing the system
- **Authentication:**
  The process of verifying the identity of a user
- **Authorization:**
  The process of granting or restricting access to system resources
- **Session:**
  A period during which a user interacts with the system after successful login

1.4 References

The development of this SRS and the SLMS system is guided by established standards and best practices to ensure quality, security, and reliability.

- **IEEE 830 Software Requirements Specification Standard**
  Provides guidelines for writing clear and structured requirement specifications
- **OWASP Web Security Guidelines**
  Offers best practices for securing web applications against common vulnerabilities
- **ISO/IEC 27001 Information Security Standards**
  Defines requirements for establishing and maintaining an information security management system
- **Web Application Security Best Practices**
  General principles for designing secure and robust web-based systems

These references ensure that the system aligns with internationally recognized standards in software engineering and cybersecurity.

1.5 Overview of Document

This document is organized into structured sections to provide a comprehensive understanding of the Student Login Management System.

- The **Introduction** section provides background information, purpose, and scope
- The **Overall Description** outlines system context, user characteristics, and assumptions
- The **System Features** section describes detailed functional requirements

- The **External Interface Requirements** define interactions with users and other systems
- The **Non-Functional Requirements** cover performance, security, usability, and reliability
- The **Database Design Considerations** describe data structures and storage requirements
- The **Security Requirements** highlight measures for protecting user data and system integrity
- The **Future Enhancements** section outlines potential improvements and scalability options

This structured approach ensures that all aspects of the system are clearly documented, making it easier for stakeholders to understand, develop, and maintain the system effectively.

2. Overall Description

The Overall Description section provides a high-level view of the Student Login Management System (SLMS). It outlines how the system fits within a broader environment, describes its major functionalities, identifies user categories, and defines the operating conditions and constraints under which the system will function.

This section serves as a conceptual foundation for understanding the system before delving into detailed functional and technical requirements.

2.1 Product Perspective

The Student Login Management System (SLMS) is designed as a **web-based authentication and access control module** that can function either as a standalone application or as an integrated component within a larger academic system.

From a system architecture perspective, SLMS operates as an intermediary layer between users and protected academic resources. It ensures that only authenticated and authorized users can gain access to the system.

The system can be integrated with:

- Academic portals
- Learning Management Systems (LMS)

- Student Information Systems (SIS)
- Other institutional platforms requiring user authentication

System Interaction Components

The SLMS interacts with several external components to function effectively:

- **Web Browsers (Client Layer):**
  Students and administrators access the system through standard web browsers, which provide the user interface for interaction.
- **Backend Application Servers:**
  These servers process business logic, handle authentication requests, manage sessions, and communicate with the database.
- **Database Management System (DBMS):**
  Stores user credentials, profiles, login records, and system logs in a structured format.
- **Email Servers:**
  Used for sending verification emails, password reset links, and system notifications.

System Architecture Overview

The SLMS follows a typical **client-server architecture**, where:

- The client (browser) sends requests
- The server processes and validates requests
- The database stores and retrieves data
- External services (e.g., email) support communication

This modular design enhances scalability, maintainability, and integration flexibility.

2.2 Product Functions

The Student Login Management System provides a comprehensive set of functions focused on authentication, user management, and system security.

Core Functionalities

- **Student Registration**
  Allows new users to create accounts by providing required personal and academic details. The system validates inputs before storing user data.

- **Secure Login Authentication**
  Verifies user identity using credentials such as email and password. Authentication ensures only legitimate users can access the system.

- **Password Reset and Recovery**
  Enables users to recover access to their accounts through secure mechanisms such as email-based reset links or OTP verification.

- **Email Verification**
  Confirms the authenticity of user accounts during registration by sending verification links or codes to registered email addresses.

- **User Session Management**
  Controls user sessions, including login duration, inactivity timeout, and secure logout to prevent unauthorized access.

- **Account Profile Management**
  Allows users to view and update their personal information such as name, email, and password.

- **Login Activity Monitoring**
  Tracks login attempts, timestamps, IP addresses (optional), and system access patterns for auditing and security purposes.

- **Account Lock Mechanism**
  Automatically locks user accounts after multiple failed login attempts to prevent brute-force attacks.

- **Administrative Account Management**
  Provides administrators with tools to manage users, enforce policies, and maintain system integrity.

These functions collectively ensure a secure, reliable, and user-friendly authentication system.

2.3 User Classes and Characteristics

The SLMS is designed to support different categories of users, each with specific roles, capabilities, and access levels.

2.3.1 Students

Students are the **primary users** of the system. They interact with the system to gain access to academic resources.

Characteristics:

- Possess basic computer literacy

- Access the system through web browsers
- Use login credentials (email and password) for authentication
- May access the system from different devices (desktop, mobile, tablet)

Capabilities:

- Register a new account
- Log in securely to the system
- Reset forgotten passwords
- Update personal profile information
- Log out securely
- Receive and respond to system notifications

2.3.2 System Administrator

The System Administrator is responsible for managing the system and ensuring its proper functioning.

Characteristics:

- Has advanced system knowledge
- Possesses elevated privileges
- Responsible for system security and maintenance

Capabilities:

- Manage student accounts (create, update, delete)
- Monitor login activities and system usage
- Lock or unlock user accounts
- Reset user passwords
- View system logs and audit trails
- Enforce security policies and access controls

2.4 Operating Environment

The SLMS is designed to operate in a modern web-based environment, supporting a wide range of devices and technologies.

Hardware Requirements

The system can run on various hardware platforms, including:

- Desktop computers
- Laptop computers
- Tablets
- Smartphones
- Web servers (for hosting backend services)

Software Requirements

Frontend Technologies:

- HTML (structure of web pages)
- CSS (styling and layout)
- JavaScript (client-side interactivity)
- React (optional, for dynamic user interfaces)

Backend Technologies:

- PHP, Node.js, or Python (for server-side logic and processing)

Database Systems:

- MySQL or PostgreSQL (for structured data storage)

Web Servers:

- Apache or Nginx (for hosting and serving the application)

Supported Web Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

The system is designed to be cross-platform compatible and responsive across devices.

2.5 Design and Implementation Constraints

The development of the SLMS must adhere to certain constraints to ensure security, compatibility, and compliance with standards.

- The system must be accessible through standard web browsers
- Passwords must be securely hashed before being stored in the database
- Secure communication protocols such as **HTTPS** must be implemented
- The system must comply with data protection and privacy regulations
- The design must support scalability and future enhancements
- The system should maintain high availability and reliability

These constraints guide the technical and architectural decisions during development.

2.6 Assumptions and Dependencies

This section outlines assumptions made during system design and external dependencies required for system operation.

Assumptions:

- Students have access to valid and active email addresses
- Users have access to internet connectivity
- The institution provides the necessary infrastructure for deployment
- Users possess basic digital literacy skills

Dependencies:

- **Database Management Systems** for storing user data
- **Email Service Providers** for sending verification and recovery messages
- **Web Hosting Infrastructure** for deploying the application
- **Network Availability** for system accessibility
- **Third-party libraries or frameworks** used in development

3. System Features

This section provides a detailed description of the core features of the **Student Login Management System (SLMS)**. Each feature is described in terms of its purpose, behaviour, and specific functional requirements. These features collectively define how the system operates and how users interact with it.

3.1 Student Registration

Description

The Student Registration feature enables new users to create accounts within the system. This is the first step in granting students access to the academic platform. The system ensures that all provided information is valid, secure, and uniquely associated with each student.

The registration process includes input validation, secure password handling, and account verification through email confirmation.

Functional Requirements

- The system must provide a **registration form** accessible to new users
- Students must provide the following required details:
  - Full Name
  - Student ID
  - Email Address
  - Password
  - Confirm Password
- The system must validate all user inputs, including:
  - Ensuring no required field is left empty
  - Validating email format
  - Ensuring Student ID uniqueness
- The system must enforce password security policies, such as:
  - Minimum length requirements (e.g., at least 8 characters)
  - Inclusion of uppercase letters, lowercase letters, numbers, and special characters
- The system must verify that the **password and confirm password fields match**
- The system must securely hash passwords before storing them in the database
- The system must store all valid user data in the database
- The system must send an **email verification link** to the registered email address
- The system must prevent duplicate account creation using the same email or student ID

3.2 Student Login

Description

The Student Login feature allows registered users to authenticate themselves and gain access to the system. This process ensures that only authorized users can access protected resources within the academic platform.

Functional Requirements

- The system must provide a **login interface** with fields for:
  - Email Address
  - Password
- The system must validate that both fields are completed before submission
- The system must verify the user's credentials against stored records in the database
- The system must compare passwords using **secure hashing algorithms** (e.g., bcrypt)
- The system must deny access if:
  - Credentials are incorrect
  - The account is not verified
  - The account is locked
- Upon successful authentication:
  - The system must create a **secure user session**
  - The user must be redirected to the **student dashboard**
- The system must display appropriate error messages for failed login attempts

3.3 Password Recovery

Description

The Password Recovery feature enables users who have forgotten their passwords to regain access to their accounts in a secure manner. This process ensures that only authorized users can reset their credentials.

Functional Requirements

- The system must provide a **"Forgot Password"** option on the login page
- The system must prompt users to enter their registered email address
- The system must validate that the email exists in the database

- The system must generate a **secure, time-limited password reset link or token**
- The system must send the reset link to the user's email address
- The system must allow users to enter and confirm a new password
- The system must enforce password security policies for the new password
- The system must securely hash and store the new password
- The system must invalidate the reset token after use or expiration

3.4 Email Verification

Description

The Email Verification feature ensures that users provide valid and accessible email addresses during registration. This process enhances system security and prevents unauthorized or fake account creation.

Functional Requirements

- The system must automatically send a **verification email** upon successful registration
- The email must contain a **unique verification link or code**
- The system must mark accounts as **unverified** until confirmation is completed
- The user must click the verification link to activate their account
- The system must update the account status to **verified** upon successful confirmation
- The system must prevent login attempts for unverified accounts
- The system must allow users to request a **new verification link** if needed

3.5 Session Management

Description

The Session Management feature controls user sessions after successful authentication. It ensures secure interaction between the user and the system while preventing unauthorized access.

Functional Requirements

- The system must create a **session** upon successful login
- The system must assign a **secure session token** to each user

- The system must maintain session state across user activities
- The system must automatically **expire sessions after a period of inactivity**
- The system must provide a **logout option** to terminate the session
- Upon logout:
    - The session must be destroyed
    - The user must be redirected to the login page
- The system must prevent session hijacking through secure practices

3.6 Account Security

Description

The Account Security feature ensures that the system protects user data and prevents unauthorized access. It implements multiple layers of security to safeguard authentication processes.

Functional Requirements

The system must implement the following security measures:

- **Password Hashing:**
  All passwords must be stored using secure hashing algorithms
- **Login Attempt Limitations:**
  The system must track failed login attempts
- **Account Lock Mechanism:**
  Accounts must be temporarily locked after a defined number of failed attempts
- **Secure Session Tokens:**
  Sessions must use randomly generated, secure tokens
- **HTTPS Encryption:**
  All data transmission must occur over secure HTTPS connections
- **Input Validation and Sanitization:**
  Prevent SQL injection, XSS, and other attacks
- **Audit Logging:**
  Record login attempts and security-related events

3.7 Profile Management

Description

The Profile Management feature allows students to view and update their personal information within the system. It ensures that user data remains accurate and up to date.

Functional Requirements

Students must be able to:

- View their profile information, including:
    - Full name
    - Email address
    - Student ID
- Update personal details (except restricted fields like Student ID if necessary)
- Change their password securely by:
    - Entering the current password
    - Providing a new password and confirming it
- The system must validate all updates before saving changes
- The system must securely store updated information

3.8 Admin Management

Description

The Admin Management feature provides administrators with the tools required to oversee system operations, manage user accounts, and maintain system security.

Functional Requirements

Administrators must be able to:

- View all registered student accounts
- Activate or deactivate user accounts
- Lock or unlock accounts based on security or administrative needs
- Reset student passwords when necessary
- Monitor login activity and system usage
- Access system logs and audit trails
- Enforce system policies and security rules

4. External Interface Requirements

This section describes how the **Student Login Management System (SLMS)** interacts with external components, including users, hardware devices, software systems, and communication protocols. These interfaces define how information is exchanged between the system and its environment.

4.1 User Interface

Overview

The User Interface (UI) defines how users interact with the system. The SLMS provides a web-based interface that is intuitive, consistent, and easy to navigate for both students and administrators.

The design of the interface prioritizes usability, accessibility, and responsiveness to ensure a seamless user experience across different devices.

System Pages

The system must provide the following key interface pages:

- **Student Login Page**
  Allows registered users to enter their credentials (email and password) to access the system.
  Features include input validation, error messages, and a "Forgot Password" option.

- **Student Registration Page**
  Enables new users to create accounts by entering required personal and academic details.
  Includes form validation and submission controls.

- **Password Reset Page**
  Allows users to reset their passwords securely using a reset link or token.
  Includes fields for new password and password confirmation.

- **Student Dashboard**
  Serves as the main interface after login.
  Displays user-specific information and provides access to system features.

- **Administrator Dashboard**
  Provides administrative controls such as user management, system monitoring, and access to logs.

User Interface Requirements

The interface must meet the following requirements:

- **User-Friendly Design:**

    - Clear navigation menus
    - Simple and intuitive layout
    - Minimal learning curve for new users
- **Responsive Design:**
    - Adaptable to different screen sizes (desktop, tablet, mobile)
    - Optimized layouts for various resolutions
- **Accessibility:**
    - Readable fonts and appropriate color contrast
    - Support for assistive technologies where possible
    - Logical tab navigation and input focus handling
- **Consistency:**
    - Uniform design elements (buttons, forms, colors) across all pages
    - Standardized feedback messages and alerts
- **Error Handling:**
    - Display meaningful error messages
    - Guide users to correct mistakes

4.2 Hardware Interface

Overview

The SLMS interacts with various hardware components that support both user access and system hosting. These interactions are indirect, as the system is web-based and accessed through browsers.

Hardware Components

- **User Devices:**
    - Desktop computers
    - Laptops
    - Tablets
    - Smartphones

These devices are used by students and administrators to access the system via web browsers.

- **Application Servers:**
  - Host the backend logic of the system
  - Process authentication requests
  - Handle session management and business logic
- **Database Servers:**
  - Store user data, credentials, and system logs
  - Ensure secure and efficient data retrieval

Hardware Interface Requirements

- The system must support **cross-device compatibility**
- The system must operate efficiently on **low and high-performance devices**
- Servers must be capable of handling **multiple concurrent users**
- Hardware must support **secure data storage and processing**

4.3 Software Interface

Overview

The SLMS integrates with various external software components to perform its operations. These interfaces ensure smooth interaction between the system and supporting technologies.

Database Systems

- **MySQL**
  Used for storing structured data such as:
  - User credentials
  - Student profiles
  - Login records
  - System logs

Requirements:

- Must support secure data storage

- Must allow efficient querying and indexing
- Must ensure data integrity and consistency

Email Systems

- **SMTP (Simple Mail Transfer Protocol) Services**
  Used for sending:
    - Email verification links
    - Password reset links
    - System notifications

Requirements:

- Reliable email delivery
- Secure transmission of email data
- Support for automated email generation

Web Servers

- **Apache**
- **Nginx**

These servers are responsible for:

- Hosting the web application
- Handling HTTP/HTTPS requests
- Serving frontend and backend resources

Requirements:

- High availability and performance
- Support for secure connections (SSL/TLS)
- Compatibility with backend technologies

4.4 Communication Interface

Overview

The Communication Interface defines how data is transmitted between system components, ensuring secure and efficient information exchange.

Communication Protocols

- **HTTPS (Hypertext Transfer Protocol Secure):**
    - All communication between clients and servers must be encrypted
    - Protects sensitive data such as login credentials
    - Prevents unauthorized interception (e.g., man-in-the-middle attacks)

Application Communication

- **REST APIs (Representational State Transfer):**
  Used for communication between:
    - Frontend (client-side) and backend (server-side)
    - External systems (if integrated)

Requirements:

- Use standard HTTP methods (GET, POST, PUT, DELETE)
- Return structured responses (e.g., JSON format)
- Implement authentication and authorization mechanisms
- Ensure secure API endpoints

Communication Requirements

- All data exchanges must be **encrypted and secure**
- The system must ensure **data integrity during transmission**
- Communication must be **fast and reliable**
- The system must handle **network failures gracefully**

5. System Requirements

This section defines the requirements that the **Student Login Management System (SLMS)** must satisfy. These requirements are categorized into **Functional Requirements**, which describe what the system should do, and **Non-Functional Requirements**, which define how the system should perform and operate under various conditions.

5.1 Functional Requirements

Functional requirements describe the core operations and behaviours of the system. These requirements specify the features and services that the SLMS must provide to users and administrators.

1. Student Registration

- The system must allow new students to create accounts
- The system must collect required user details (name, student ID, email, password)
- The system must validate all input fields before submission
- The system must prevent duplicate accounts using the same email or student ID
- The system must store user data securely in the database
- The system must send a verification email upon successful registration

2. Student Login Authentication

- The system must allow registered users to log in using email and password
- The system must validate user credentials against stored records
- The system must use secure password hashing for verification
- The system must deny access for invalid credentials or unverified accounts
- The system must create a session upon successful login
- The system must redirect users to the dashboard after login

3. Password Reset

- The system must provide a “Forgot Password” feature
- The system must allow users to request a password reset using their email
- The system must send a secure, time-limited reset link
- The system must allow users to create a new password
- The system must enforce password security rules during reset
- The system must securely update the password in the database

4. Email Verification

- The system must send a verification link during registration

- The system must require users to verify their email before login
- The system must activate accounts only after successful verification
- The system must allow resending of verification emails

5. Session Management

- The system must create a session after successful authentication
- The system must maintain session state securely
- The system must automatically log out users after inactivity
- The system must provide a logout function
- The system must destroy session data upon logout

6. Administrative Control

- The system must provide an admin interface/dashboard
- Administrators must be able to:
    - View all student accounts
    - Activate or deactivate accounts
    - Reset user passwords
    - Monitor login activity
    - Lock or unlock accounts

5.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system, including performance, reliability, usability, and scalability. These requirements ensure that the system operates efficiently and meets user expectations.

5.2.1 Performance

The system must deliver fast and efficient responses under normal and peak conditions.

- The system must support **at least 500 concurrent users** without performance degradation
- The system must process login and authentication requests within **2–3 seconds**
- The system must handle database queries efficiently
- The system must minimize page load times

- The system must optimize resource usage (CPU, memory, bandwidth)

5.2.2 Reliability

The system must operate consistently and be available when needed.

- The system should maintain **high availability (e.g., 99% uptime or higher)**
- The system must handle failures gracefully without data loss
- The system must include error handling and recovery mechanisms
- The system must maintain data integrity during transactions
- The system must ensure consistent performance over time

5.2.3 Usability

The system must be easy to learn and use for all users, regardless of technical skill level.

- The interface must be **clear, simple, and intuitive**
- Navigation must be logical and consistent across all pages
- The system must provide helpful error messages and user guidance
- The system must minimize the number of steps required to complete tasks
- The system must support accessibility best practices (readability, responsiveness)

6. Database Requirements

Example Database Tables

Students Table

• student_id

• name

• email

• password_hash

• email_verified

• created_at

Login Logs Table

• log_id

• student_id

• login_time

• ip_address

• status

Password Reset Table

• reset_id

• student_id

• reset_token

• expiry_time

7. Security Requirements

The system must implement strong security mechanisms.

Security measures include:

• Password hashing using secure algorithms

• Protection against SQL injection

• Cross-site scripting (XSS) prevention

• Cross-site request forgery (CSRF) protection

• Secure session management

• HTTPS communication

Authentication must be properly enforced to ensure only authorized users access the system.

8. System Constraints

The system must operate within the following constraints:

- Requires internet connectivity
- Must run on modern web browsers
- Must use secure authentication methods
- Must comply with institutional data policies

9. Future Enhancements

Possible improvements for the system include:

- Two-factor authentication (2FA)
- Biometric authentication
- Mobile application integration
- Single Sign-On (SSO)
- Integration with learning management systems